Where to Predict in Retinal OCT? Anatomy-Guided Target Selection for I-JEPA

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paper, code & pre-registration

Why target selection matters in OCT

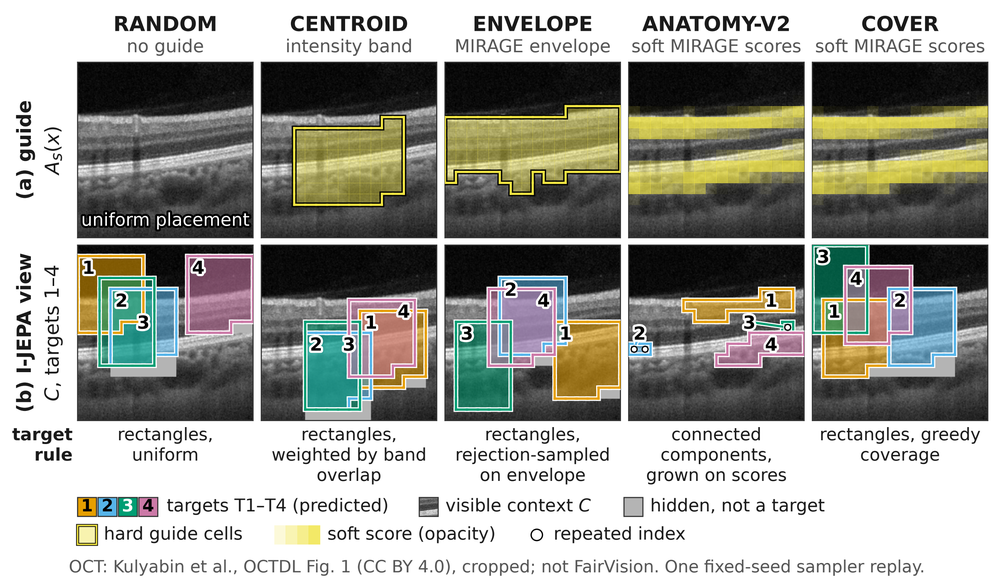
- In retinal OCT the layered retina fills only a band of each B-scan; much of the rest is near-uniform background. Near-uniform pixels can still contribute informative latent context through attention.
- I-JEPA learns by predicting the *latent features* of hidden target blocks from visible context, but places targets uniformly.

Question. Does steering I-JEPA targets toward retinal tissue give a better frozen encoder for glaucoma detection, with the same architecture and latent loss?

Answer (original single runs). Tissue-directed rectangles scored higher than uniform I-JEPA masking at epoch 50: +0.0099 AUC with the segmentation-free CENTROID, +0.0120 with ENVELOPE. More anatomical detail did not add gains; CENTROID and ENVELOPE also leave more context visible.

Registered replication: CENTROID *incomplete*, ENVELOPE *incomplete*.

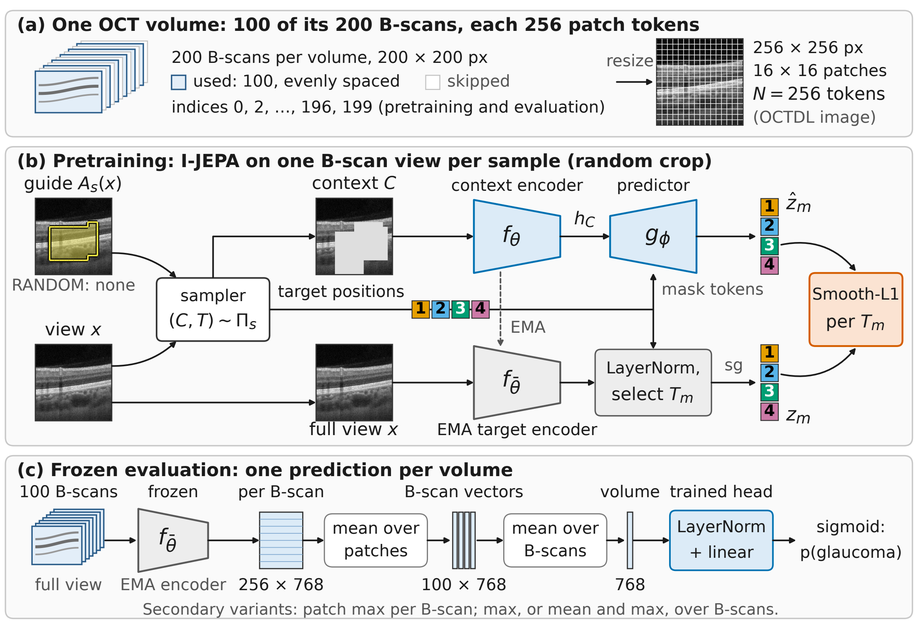
Five target-selection strategies



OCT: Kulyabin et al., OCTDL Fig. 1 (CC BY 4.0), cropped; not FairVision. One fixed-seed sampler replay.

RANDOM targets cover much background; guided rectangles cluster on the tissue band; ANATOMY-V2 targets are thin tissue-shaped components.

Same I-JEPA, different sampler; volume-level probe



Only the sampler changes: $(C, T) \sim \Pi_S(\cdot \mid A_S(x))$ with the guide $A_S(x)$; encoder, EMA teacher, predictor and Smooth-L1 loss are shared.

Single runs: tissue-directed rectangles scored higher

FairVision glaucoma OCT; frozen EMA encoder, mean-pooled features, linear probe; test N = 3000 volumes (1466 glaucoma). All continuations start from one epoch-25 uniform I-JEPA checkpoint.

	epoch 50		epoch 75		epoch 100	
strategy	AUC	Δ	AUC	Δ	AUC	Δ
RANDOM	0.8641	—	0.8723	—	0.8746	—
ENVELOPE	0.8761	+0.0120	0.8803	+0.0080	0.8807	+0.0062
CENTROID	0.8740	+0.0099	0.8836	+0.0113	0.8855	+0.0109
ANATOMY-V2	0.8654	+0.0013	0.8612	−0.0111	—	—
COVER	0.8643	+0.0002	0.8639	−0.0084	0.8577	−0.0168

Δ: minus RANDOM at the same epoch; upper rows fp16 probes, lower rows fp32 (vs an fp32 RANDOM); ANATOMY-V2 stopped at epoch 75. One continuation per strategy: intervals reflect test-case sampling (10,000 paired bootstrap resamples), not retraining.

At epoch 50 (registered endpoint): ENVELOPE +0.0120 AUC (95% CI [+0.0068, +0.0173]), CENTROID (no segmenter) +0.0099 (95% CI [+0.0051, +0.0149]); positive at epochs 50, 75 and 100 (ENVELOPE +0.0062 to +0.0120, CENTROID +0.0099 to +0.0113).
More anatomy did not add gains: ANATOMY-V2 minus ENVELOPE −0.0107 at epoch 50; COVER minus RANDOM +0.0002 at epoch 50, −0.0168 at epoch 100.

Which masks actually reach the model?

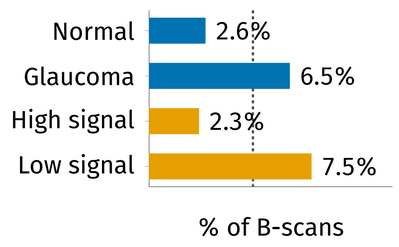
Sampler replay in each trained configuration: 6,400 training views, batch 64, 3 sampler seeds. Tissue is a MIRAGE-derived proxy, not ground truth.

strategy	tissue hidden	target purity	unique targets	loss slots	context (% grid)	AUC ep. 50
RANDOM	53.2%	30.9%	113.1	160.2	27.5 ± 0.1	0.8641
CENTROID	59.7%	38.2%	102.5	160.2	31.5 ± 0.4	0.8740
ENVELOPE	70.7%	42.1%	110.2	160.2	29.2 ± 0.5	0.8761
COVER	73.3%	40.1%	120.1	160.2	25.2 ± 0.4	0.8643
ANATOMY-V2	79.7%	96.9%	54.0	64.0	63.3 ± 0.4	0.8654

- CENTROID and ENVELOPE leave more context visible than RANDOM: +4.0 points (95% CI 3.6–4.4) and +1.6 (1.1–2.1); COVER leaves less.
- ANATOMY-V2 has the purest targets (96.9%) but 64.0 loss slots versus 160.2.

Target location, target size and visible context are coupled.

How reliable are the guides?



CENTROID band vs MIRAGE envelope: gross disagreement in 4.8% of B-scans (dashed; 95% CI 3.7–5.9; all training volumes 5.3%).

Envelope plausibility flags: 3.8% (2.7–5.0).

MIRAGE guide: GOALS foreground Dice 0.925, in-domain only.

Agreement of two automatic guides, not anatomical accuracy (no layer labels). Bars: unweighted rates, stratified sample of 300 volumes (3,000 B-scans); signal: tertiles of an SNR proxy.

Takeaways

- Tissue-directed rectangles improved frozen glaucoma probes in the original runs, even with a segmentation-free intensity centroid.
- More anatomical specificity (tissue shapes, coverage) added no further gain.
- Target placement, target size and visible context move together. Registered replication labels: CENTROID *incomplete*, ENVELOPE *incomplete*.

Registered replication: repeated continuations and matched-budget control

Repeated continuations Epoch-50 test AUC; originals re-probed, new runs from the epoch-25 checkpoint. Mean and Δ (vs RANDOM, same seed): registered seeds.

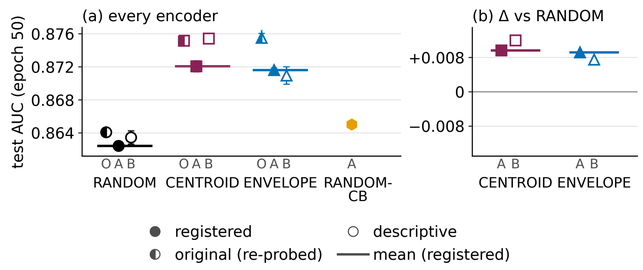
strategy	original	seed A	seed B*	mean	mean Δ
RANDOM	0.8641	0.8624	0.8635	0.8624	—
ENVELOPE	0.8755	0.8716	0.8710	0.8716	+0.0092
CENTROID	0.8752	0.8721	0.8754	0.8721	+0.0096

Matched-budget control RANDOM-CB: uniform placement with CENTROID's per-image budgets; seed A, one run per arm.

contrast	Δ AUC
CENTROID − RANDOM-CB (placement)	+0.0070
RANDOM-CB − RANDOM (budget)	+0.0026

Pooling Maxima within or across B-scans; mean Δ vs RANDOM (descriptive; mean+max: paper).

max	within	across
ENVELOPE	+0.0069	+0.0184
CENTROID	+0.0142	+0.0188



Error bars: s.d. over probe-head seeds. Letters: seed index (O: original). Open markers: descriptive runs (late or incomplete). (b): guided minus RANDOM, same seed.

Registered labels: CENTROID *incomplete*, ENVELOPE *incomplete*. **Matched-budget control:** not assessed (no registered RANDOM range). *Incomplete at the freeze: descriptive. **Label rule:** consistent direction if every registered Δ > 0 and mean Δ > RANDOM range; reversed if mean Δ ≤ 0; else within run-to-run variation; incomplete with too few on-time runs.

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I-JEPA: Assran et al., CVPR 2023. MIRAGE: Morano et al., npj Digit. Med. 2025. FairVision: Luo et al., 2024. OCT example in both OCT figures: Kulyabin et al., OCTDL, Fig. 1 (arXiv:2312.08255v3), adapted (cropped, downsampled, overlays added) under CC BY 4.0 (creativecommons.org/licenses/by/4.0); illustration only. No FairVision images are shown.